

## SEOSAN, Korea, Republic of

WMO: 471290

Lat: 36.7767N

Lon: 126.4939E

Elev: 99

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 14.0       | 16.9 | 2.6                           | 6.3 | 21.7 | 5.6 | 7.4 | 23.0 | 21.1                  | 29.0 | 18.3 | 29.7 | 4.5                      | 50   | 0.423 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.3                   | 89.4            | 77.8 | 86.9 | 76.7 | 84.5 | 75.1 | 80.6                | 85.2 | 79.0 | 83.9 | 77.7 | 82.5 | 6.9                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.5                            | 153.4 | 82.9 | 77.8 | 144.8 | 81.5 | 76.3 | 137.9 | 80.6 | 44.4          | 85.1 | 42.7 | 84.0 | 41.4 | 82.8 | 87.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |
| 18.9              | 16.2 | 14.1 | DB                         | 8.6 | 92.7               | 3.8 | 2.5                                                | 5.9 | 94.5       | 3.6 | 96.0       | 1.5 | 97.5       | -1.3 | 99.3 |
|                   |      |      | WB                         | 7.6 | 81.5               | 3.9 | 2.6                                                | 4.8 | 83.4       | 2.6 | 84.9       | 0.4 | 86.3       | -2.4 | 88.2 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 54.3  | 29.2  | 32.4  | 40.8  | 51.9  | 62.3  | 70.8  | 76.9  | 78.2  | 69.8  | 58.4  | 45.7  | 33.7 | (6)  |
| (7)  |                                                                   | DBStd       | 17.90 | 6.36  | 6.40  | 6.09  | 6.32  | 4.71  | 3.43  | 3.73  | 4.05  | 4.54  | 5.87  | 7.06  | 6.69 | (7)  |
| (8)  |                                                                   | HDD50       | 2163  | 646   | 493   | 292   | 52    | 0     | 0     | 0     | 0     | 9     | 165   | 506   |      | (8)  |
| (9)  |                                                                   | HDD65       | 5056  | 1111  | 913   | 750   | 394   | 110   | 3     | 0     | 11    | 214   | 579   | 971   |      | (9)  |
| (10) |                                                                   | CDD50       | 3730  | 0     | 1     | 7     | 109   | 381   | 625   | 833   | 874   | 594   | 270   | 36    | 0    | (10) |
| (11) |                                                                   | CDD65       | 1147  | 0     | 0     | 0     | 1     | 26    | 178   | 368   | 409   | 155   | 10    | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 7925  | 0     | 0     | 0     | 15    | 169   | 839   | 2594  | 3486  | 792   | 30    | 0     | 0    | (12) |
| (13) |                                                                   | CDH80       | 1997  | 0     | 0     | 0     | 1     | 14    | 123   | 658   | 1084  | 117   | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 5.4   | 5.2   | 5.5   | 6.6   | 6.7   | 6.3   | 5.1   | 6.0   | 5.2   | 4.4   | 4.2   | 4.9   | 5.1  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 47.2  | 1.2   | 1.3   | 1.8   | 3.3   | 3.8   | 4.4   | 10.3  | 9.3   | 5.8   | 2.4   | 2.2   | 1.3  | (15) |
| (16) |                                                                   | PrecMax     | 70.4  | 3.5   | 3.5   | 4.6   | 8.3   | 8.8   | 17.3  | 20.0  | 21.0  | 14.8  | 11.5  | 6.5   | 3.1  | (16) |
| (17) |                                                                   | PrecMin     | 27.0  | 0.1   | 0.0   | 0.1   | 0.3   | 0.3   | 0.4   | 2.3   | 3.3   | 0.6   | 0.0   | 0.5   | 0.3  | (17) |
| (18) |                                                                   | PrecStd     | 9.5   | 0.9   | 0.9   | 1.1   | 2.1   | 2.2   | 3.9   | 4.1   | 4.6   | 3.6   | 2.8   | 1.6   | 0.7  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 50.3  | 57.0  | 63.9  | 76.1  | 81.9  | 86.2  | 93.4  | 93.3  | 85.9  | 77.1  | 67.7  | 55.3 | (19) |
| (20) |                                                                   | 0.4%        | MCWB  | 45.2  | 48.8  | 52.2  | 58.3  | 63.4  | 68.8  | 77.6  | 79.7  | 72.5  | 64.6  | 57.7  | 48.3 | (20) |
| (21) |                                                                   | 2%          | DB    | 45.2  | 50.7  | 59.2  | 70.9  | 78.2  | 83.3  | 88.8  | 90.4  | 83.3  | 74.3  | 63.6  | 50.9 | (21) |
| (22) |                                                                   | 2%          | MCWB  | 39.9  | 43.6  | 49.0  | 56.7  | 62.2  | 68.3  | 77.7  | 78.6  | 71.0  | 63.2  | 56.5  | 45.1 | (22) |
| (23) |                                                                   | 5%          | DB    | 42.0  | 46.8  | 55.7  | 67.6  | 75.4  | 81.0  | 86.1  | 88.1  | 80.9  | 71.6  | 60.7  | 47.6 | (23) |
| (24) |                                                                   | 5%          | MCWB  | 37.3  | 40.5  | 47.0  | 54.7  | 61.3  | 68.3  | 77.3  | 77.8  | 70.3  | 61.6  | 53.8  | 42.8 | (24) |
| (25) |                                                                   | 10%         | DB    | 39.2  | 43.5  | 52.3  | 64.2  | 72.6  | 78.9  | 83.5  | 85.8  | 78.7  | 68.9  | 57.6  | 44.6 | (25) |
| (26) |                                                                   | 10%         | MCWB  | 35.0  | 38.1  | 44.7  | 53.4  | 60.6  | 68.0  | 76.0  | 76.8  | 68.9  | 60.1  | 51.7  | 40.1 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 46.3  | 51.8  | 56.5  | 63.4  | 68.7  | 74.6  | 81.9  | 83.5  | 77.1  | 69.6  | 61.8  | 50.3 | (27) |
| (28) |                                                                   | 0.4%        | MCDWB | 49.0  | 53.1  | 60.5  | 69.9  | 73.9  | 79.8  | 85.7  | 88.0  | 81.5  | 73.9  | 65.2  | 53.7 | (28) |
| (29) |                                                                   | 2%          | WB    | 41.1  | 44.7  | 52.0  | 60.1  | 66.4  | 72.8  | 79.9  | 81.4  | 75.0  | 66.4  | 58.4  | 46.4 | (29) |
| (30) |                                                                   | 2%          | MCDWB | 44.4  | 49.0  | 56.5  | 66.7  | 72.6  | 77.9  | 84.4  | 86.1  | 79.7  | 70.6  | 62.1  | 49.8 | (30) |
| (31) |                                                                   | 5%          | WB    | 38.1  | 41.2  | 48.4  | 57.3  | 64.4  | 71.5  | 78.6  | 79.9  | 73.3  | 64.1  | 55.3  | 43.2 | (31) |
| (32) |                                                                   | 5%          | MCDWB | 41.0  | 45.8  | 53.4  | 64.1  | 71.2  | 76.7  | 83.2  | 84.8  | 77.3  | 68.7  | 58.7  | 47.0 | (32) |
| (33) |                                                                   | 10%         | WB    | 35.5  | 38.6  | 45.6  | 54.9  | 62.7  | 70.2  | 77.4  | 78.6  | 71.5  | 62.0  | 52.6  | 40.6 | (33) |
| (34) |                                                                   | 10%         | MCDWB | 38.9  | 42.9  | 51.0  | 61.6  | 69.3  | 75.4  | 82.1  | 83.6  | 75.8  | 66.9  | 56.7  | 44.5 | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 14.7  | 16.7  | 18.3  | 19.5  | 17.5  | 14.6  | 10.8  | 12.3  | 15.8  | 18.7  | 16.7  | 14.7 | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 18.1  | 21.7  | 23.3  | 24.2  | 22.5  | 19.1  | 14.9  | 15.6  | 17.9  | 20.5  | 20.3  | 18.7 | (36) |
| (37) |                                                                   | 5% DB       | MCWBR | 13.3  | 14.9  | 14.2  | 12.1  | 9.5   | 7.5   | 6.3   | 6.2   | 7.8   | 10.5  | 12.9  | 13.7 | (37) |
| (38) |                                                                   | 5% WB       | MCDBR | 16.1  | 19.2  | 19.1  | 18.7  | 17.0  | 13.4  | 11.8  | 12.7  | 13.2  | 16.9  | 16.3  | 17.4 | (38) |
| (39) |                                                                   | 5% WB       | MCWBR | 13.2  | 14.8  | 13.7  | 11.5  | 9.5   | 6.8   | 6.1   | 6.7   | 7.6   | 10.9  | 12.7  | 14.0 | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.447 | 0.491 | 0.569 | 0.600 | 0.619 | 0.692 | 0.650 | 0.577 | 0.509 | 0.497 | 0.479 | 0.447 | (40) |      |
| (41) |                                                                   | taud        | 1.937 | 1.861 | 1.721 | 1.700 | 1.714 | 1.638 | 1.775 | 1.953 | 2.060 | 1.955 | 1.942 | 1.953 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 220   | 227   | 222   | 225   | 223   | 207   | 215   | 228   | 237   | 223   | 207   | 209   | (42) |      |
| (43) |                                                                   | Edh at Noon | 45    | 55    | 69    | 75    | 75    | 81    | 70    | 57    | 49    | 49    | 45    | 42    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 768   | 1032  | 1396  | 1624  | 1797  | 1662  | 1374  | 1448  | 1343  | 1161  | 780   | 667   | (44) |      |
| (45) |                                                                   | RadStd      | 50    | 90    | 137   | 119   | 177   | 178   | 208   | 211   | 139   | 87    | 84    | 48    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (42 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|      |                         |       |         |        |         |       |       |             |       | +167  | +95   |

Nomenclature: See separate page